`genQE_FSI` — Two-Nucleon GCF Generator with GENIE Intranuclear Cascade

Physics & Pipeline Documentation

GCF with 3N Project

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1 Overview

`genQE_FSI` wraps the two-nucleon GCF quasi-elastic generator (`genQE`) with a **Final State Interaction (FSI)** layer. After the GCF vertex produces a correlated nucleon pair, both the struck ("lead") and partner ("recoil") nucleons are independently transported through the residual nucleus using GENIE's **Intranuke2018** intranuclear cascade (INC).

The reaction is

$$e + A \longrightarrow e' + N'_{\text{lead}} + N'_{\text{rec}} + X + (A{-}2)^*, \tag{1}$$

where primed nucleons denote post-FSI momenta and X represents any additional secondaries (pions, ejected nucleons) produced during re-scattering.

Two cascade models are available:

- **hN** (`HNIntranuke2018`): full hadron–nucleon cascade with explicit angular distributions. More accurate.
- **hA** (`HAIntranuke2018`): effective hadron–nucleus cascade using data-driven total cross sections. Faster ($\sim$5–10$\times$), suitable for systematics studies.

2 Pre-FSI Stage: GCF Event Generation

The GCF vertex is identical to `genQE` (see its `PHYSICS.tex`). It produces:

- Lead and recoil nucleon 4-momenta ($\boldsymbol{p}_{\text{lead}}$, $\boldsymbol{p}_{\text{rec}}$)
- Pair types (PDG codes)
- Scattered electron 4-momentum $\boldsymbol{k}$
- Event weight w from the GCF spectral function and e–N cross section

At this stage the nucleons have momenta but *no spatial position* — the GCF is a momentum-space formalism.

3 SRC Position Sampling

FSI requires a starting position inside the nucleus. Since the SRC pair was formed by two nucleons overlapping at short range, the probability of finding the pair at radius r scales as the *square* of the single-nucleon density:

$$P(r)\,dr \;\propto\; \rho^2(r)\;4\pi r^2\,dr. \tag{2}$$

3.1 Density models

GENIE's `genie::utils::nuclear::Density(`r`,` A`)` returns the single-nucleon density using:

Woods–Saxon (A > 20).

$$\rho_{\text{WS}}(r) = \frac{\rho_0}{1+\exp\!\big[(r-R)/a\big]}, \qquad R = r_0\,A^{1/3}, \quad a \approx 0.545\,\text{fm}, \tag{3}$$

with $r_0 \approx 1.07\,\text{fm}$ and ρ_0 set by normalization to $\rho_{\text{normal}} = 0.17\,\text{fm}^{-3}$.

Modified Gaussian ($4 < A \leq 20$).

$$\rho_{\text{G}}(r) = \rho_0\,\exp\!\left[-\left(\frac{r}{R}\right)^2\right], \qquad R = r_0\,A^{1/3}. \tag{4}$$

Parameters are fitted to electron-scattering charge-density data (de Vries *et al.*, At. Data Nucl. Data Tables **36**, 1987, doi:10.1016/0092-640X(87)90013-1).

3.2 Rejection sampling algorithm

The radial coordinate is sampled via rejection:

1. Compute $R_{\max} = 2.5 \times 1.4 \times A^{1/3}$ fm.
2. Find the envelope $f_{\max} = 1.01 \max_r \left[\rho^2(r)\, r^2\right]$ by scanning 200 grid points (cached per A).
3. Draw $r \sim \text{Uniform}(0,\, R_{\max})$.
4. Accept if $U \cdot f_{\max} \leq \rho^2(r)\, r^2$ with $U \sim \text{Uniform}(0, 1)$.

An isotropic direction $(\cos\theta,\, \phi)$ is drawn uniformly. Both SRC nucleons are placed at the *same* spatial position $\boldsymbol{x}_{\text{SRC}}$, reflecting their short-range overlap.

3.3 Caching

The envelope $f_{\max}$ and $R_{\max}$ are stored as statics and recomputed only when A changes, avoiding ~ 200 density evaluations per event.

4 GENIE Intranuclear Cascade — General Framework

The remainder of this document describes how GENIE's hN cascade works in detail. The hA model follows similar logic with data-driven effective cross sections.

4.1 Event record construction

GENIE's cascade operates on a `GHepRecord` (event record). We construct a minimal record to transport a single nucleon:

Slot	PDG	Status	Purpose
0	e^- (11)	`kIStInitialState`	Fake probe; triggers `kGMdLeptonNucleus` mode
1	Nucleus (A,Z)	`kIStInitialState`	Target; used by `SetTrackingRadius`
2	Nucleus (A,Z)	`kIStStableFinalState`	Remnant nucleus
3	Nucleon	`kIStHadronInTheNucleus`	Particle to transport; carries sampled position

Why a fake electron? GENIE determines the "event generation mode" from the PDG of slot 0. If it sees a lepton (e, μ, ν), it enters `kGMdLeptonNucleus` mode, which *skips* `GenerateVertex()` — the routine that would otherwise overwrite our sampled position with a random point on the nuclear surface.

The fake electron is set to $E = 100\,\text{GeV}$ to avoid triggering energy sanity checks in the hA model. It is never transported (only particles with status `kIStHadronInTheNucleus` are stepped).

4.2 Tracking radius

Before the cascade begins, GENIE sets a tracking sphere:

$$R_{\text{track}} = f_{NR} \times R_0 \times A^{1/3}, \qquad R_0 = 1.4\,\text{fm}, \quad f_{NR} = 3. \tag{5}$$

A hadron is considered "inside the nucleus" while $|\boldsymbol{x}| < R_{\text{track}} + \delta r$, where $\delta r = 0.05\,\text{fm}$ (one step).

4.3 Stepping

The cascade is semi-classical: the nucleon is stepped through the nucleus in fixed increments of

$$\delta r = 0.05\,\text{fm}. \tag{6}$$

At each step the particle's position is advanced along its momentum direction. A boundary check prevents it from flying more than 1 fm past the nuclear surface.

5 Mean Free Path and Interaction Probability

At each step the interaction probability is determined by the **mean free path**

$$\lambda(r) = \frac{1}{\rho(r)\,\sigma_{\text{tot}}}, \tag{7}$$

where $\rho(r)$ is the local nuclear density and σ_{tot} is the total hadron–nucleon cross section (in fm^2).

5.1 Nuclear density with de Broglie correction

The density entering Eq. (7) is the full nuclear density $A\,\rho(r)$ (i.e., number density, not single-nucleon density), with an empirical *de Broglie ring correction* that accounts for the finite size of the interaction region:

$$\rho_{\text{eff}}(r) = A\;\rho(r,\;A,\;\text{ring})\,, \qquad \text{ring} = n \times \frac{\hbar c}{p}, \tag{8}$$

where p is the hadron's momentum and n depends on A and particle type:

- Pions: $n = \delta R_\pi$ (for $A > 20$, doubled for light nuclei)
- Nucleons: $n = \delta R_N$ (similar)

Default values: $\delta R_\pi \approx \delta R_N \approx 1.5$.

5.2 Total cross section

For a **proton** at kinetic energy T:

$$\sigma^{p}_{\text{tot}} = \sigma_{pp}(T)\,\frac{Z}{A} + \sigma_{pn}(T)\,\frac{A-Z}{A} + \sigma_{\text{Coulomb}}, \tag{9}$$

where σ_{pp}, σ_{pn} are parameterized from data (tabulated splines in `INukeHadroData2018`, clamped to $T \in [10,\,1000]\,\text{MeV}$). A Coulomb correction (Cohen formula with penetrability factor) is added for protons.

An empirical low-energy cutoff sets $\sigma_{\text{tot}} = 0$ if $T < A^{0.2} \times 12\,\text{MeV}$.

For **pions** at $T \geq 350\,\text{MeV}$:

$$\sigma^{\pi}_{\text{tot}} = \sigma_{\pi p}(T)\,\frac{Z}{A} + \sigma_{\pi n}(T)\,\frac{A-Z}{A}. \tag{10}$$

For $T < 350\,\text{MeV}$ the Oset model is used instead (Section 7).

5.3 Pandharipande–Pieper medium corrections

For nucleon–nucleon scattering, the free-space cross section is modified by an in-medium correction factor:

$$\sigma^{NN,\text{med}}_{\text{tot}} = \sigma^{NN,\text{free}}_{\text{tot}} \times f_{\text{PP}}(\rho,\,T), \tag{11}$$

where $f_{\text{PP}} \in [0.3,\,0.8]$ typically, accounting for:

Pauli blocking. The fraction of NN scattering final states that are *above* the local Fermi surface. The local Fermi momentum is

$$k_{F,p} = \left(\frac{3\pi^2\,\rho\,Z}{A}\right)^{1/3}\frac{1}{\hbar c}, \qquad k_{F,n} = \left(\frac{3\pi^2\,\rho\,(A-Z)}{A}\right)^{1/3}\frac{1}{\hbar c}. \tag{12}$$

In-medium effective mass. The nucleon acquires a momentum-dependent effective mass

$$m^*(\rho,\,k^2) = m_N\,\frac{(\Lambda^2+k^2)^2}{(\Lambda^2+k^2)^2 - 2\Lambda^2\,B\,m_N}, \tag{13}$$

with $\Lambda(\rho) = \lambda_0 + \lambda_1\,\rho$ and $B(\rho) = \beta_1\,\rho$ parameterized from nuclear-matter calculations (Pandharipande & Pieper, Phys. Rev. C **45**, 791, 1992, doi:10.1103/PhysRevC.45.791).

The correction is pre-tabulated on a 200×17 grid in $(T,\,\rho)$ and bilinearly interpolated.

5.4 Interaction sampling

At each step, an interaction distance is drawn from an exponential:

$$d = -\lambda\,\ln U, \qquad U \sim \mathrm{Uniform}(0,1). \tag{14}$$

If $d < \delta r = 0.05\,\mathrm{fm}$, the hadron interacts; otherwise it continues to the next step.

6 Hadron Fates

When an interaction occurs, a **fate** is selected by drawing from the relative cross sections. Six outcomes are possible:

6.1 Elastic scattering (`kIHNFtElas`)

The hadron scatters elastically off a single bound nucleon. No particle species change; only momenta are redistributed.

Kinematics. The scattering angle $\cos\theta_{\mathrm{CM}}$ is sampled from parameterized NN angular distributions (tabulated in `INukeHadroData2018`). Full relativistic two-body kinematics are applied:

1. Boost to the CM frame of the hadron–nucleon system.
2. Compute outgoing CM momenta:
$$E_3^{\mathrm{CM}} = \frac{s + m_3^2 - m_4^2}{2\sqrt{s}}, \qquad |\boldsymbol{p}_3^{\mathrm{CM}}| = \sqrt{(E_3^{\mathrm{CM}})^2 - m_3^2}. \tag{15}$$
3. Rotate by sampled $(\theta_{\mathrm{CM}},\,\phi_{\mathrm{CM}})$ with ϕ uniform in $[0, 2\pi)$.
4. Boost back to the lab frame.

The struck nucleon's Fermi momentum is sampled from the local Fermi gas (when `fDoFermi` = true, default). The remnant nucleus 4-momentum is updated accordingly.

6.2 Charge exchange (`kIHNFtCEx`)

The hadron exchanges charge with a bound nucleon:

$$\pi^+ + n \to \pi^0 + p, \qquad \pi^- + p \to \pi^0 + n, \tag{16}$$

$$\pi^0 + p \to \pi^+ + n, \qquad \pi^0 + n \to \pi^- + p. \tag{17}$$

For nucleons, charge exchange corresponds to $p+n \to n+p$ (with different momentum sharing). The kinematics are identical to elastic scattering but with different outgoing particle masses and PDG codes.

Availability. Charge exchange requires the appropriate partner nucleon to be present in the remnant (e.g., π^+ CEx needs at least one neutron). If unavailable, this fate's weight is set to zero.

6.3 Inelastic / pion production (`kIHNFtInelas`)

At sufficiently high energy, the collision produces one or more pions:

$$N + N \to N + N + n\pi, \qquad \pi + N \to N + m\pi. \tag{18}$$

The final state is generated using GENIE's hadronization routines, which sample multiplicity and kinematics from parameterized inclusive data. Each produced pion is itself subject to further cascade steps.

6.4 Absorption (`kIHNFtAbs`)

The hadron is absorbed by a correlated nucleon pair (typically a deuteron-like pn system):

$$\pi^+ + (pn) \to p + p, \tag{19}$$
$$\pi^- + (pn) \to n + n, \tag{20}$$
$$\pi^0 + (pn) \to p + n. \tag{21}$$

This requires *both* a proton and a neutron in the remnant.

Kinematics. Two target nucleons are sampled with Fermi momenta (scaled by an empirical factor $f_{\text{Fermi}} = 0.6$). Two-body kinematics are applied in the $\pi+(NN)$ CM frame. The outgoing nucleons are Pauli-blocked: if either falls below k_F, the absorption is rejected and re-sampled.

The remnant loses two nucleons: $A_{\text{rem}} \to A_{\text{rem}} - 2$, $Z_{\text{rem}} \to Z_{\text{rem}} - \Delta Z$.

6.5 Compound nucleus (`kIHNFtCmp`)

At very low kinetic energies ($T \lesssim 30\,\text{MeV}$ for $A > 5$), a nucleon may be captured into the remnant nucleus, forming a compound system. The compound nucleus subsequently decays via pre-equilibrium emission (a fast nucleon is ejected carrying a fraction of the excitation energy) or statistical equilibrium decay.

6.6 No interaction (`kIHNFtNoInteraction`)

The hadron escapes the nucleus without any interaction. This is the outcome when $d > \delta r$ at every step until $|\boldsymbol{x}| > R_{\text{track}}$.

6.7 Fate selection

Each fate's probability is proportional to its partial cross section. For nucleons:

$$P(\text{fate}) = \frac{\sigma_{\text{fate}}(T,\, A,\, Z)}{\sigma_{\text{elas}} + \sigma_{\text{inel}} + \sigma_{\text{cmp}}}. \tag{22}$$

For pions (high energy):

$$P(\text{fate}) = \frac{\sigma_{\text{fate}}}{\sigma_{\text{CEx}} + \sigma_{\text{elas}} + \sigma_{\text{inel}} + \sigma_{\text{abs}}}. \tag{23}$$

Each partial cross section can be scaled by a tunable "fate weight" factor (e.g., `fNucAbsFac` for absorption). For π^0 absorption, an additional isospin factor of 0.665 is applied.

A cumulative distribution is formed and a uniform random number selects the fate.

7 Oset Model for Low-Energy Pions

For pion kinetic energies $T_\pi < 350\,\mathrm{MeV}$, the cascade uses the **Oset model** (Salcedo *et al.*, Nucl. Phys. A **484**, 557, 1988, doi:10.1016/0375-9474(88)90310-7), which provides density-dependent pion–nucleon cross sections dominated by the $\Delta(1232)$ resonance.

7.1 Δ propagator

The πN invariant mass is

$$M_{\mathrm{inv}} = \sqrt{m_\pi^2 + 2 m_N (T_\pi + m_N) + m_N^2}. \tag{24}$$

The Δ propagator denominator is

$$|D_\Delta|^2 = (M_{\mathrm{inv}} - m_\Delta)^2 + (\Gamma_{\mathrm{red}} + \Sigma)^2, \tag{25}$$

where Γ_{red} is the reduced (Pauli-blocked) Δ decay width and Σ is the in-medium self-energy.

7.2 Reduced width

The free-space $\Delta \to \pi N$ width at invariant mass M_{inv} is

$$\Gamma_{\mathrm{free}} = \frac{g^2}{12\pi}\, f_{\mathrm{coupl}}\, \frac{m_N\, p_{\mathrm{CM}}^3}{M_{\mathrm{inv}}}, \qquad p_{\mathrm{CM}} = \frac{p_\pi\, m_N}{M_{\mathrm{inv}}}. \tag{26}$$

This is multiplied by a **Pauli reduction factor**:

$$R_{\mathrm{Pauli}} = \begin{cases} 0 & \mu_0 < -1 \\ \dfrac{\mu_0^3 + \mu_0 + 2}{4} & -1 \le \mu_0 \le 1 \\ 1 & \mu_0 > 1 \end{cases} \tag{27}$$

with

$$\mu_0 = \frac{E_\Delta\, E_N^{\mathrm{CM}} - E_F\, M_{\mathrm{inv}}}{p_\pi\, p_{\mathrm{CM}}}, \qquad E_F = \sqrt{k_F^2 + m_N^2}, \quad k_F = \left(\frac{3\pi^2 \rho}{2}\right)^{1/3} \hbar c. \tag{28}$$

7.3 Self-energy

The in-medium Δ self-energy Σ contains absorption terms (NN and $3N$) parameterized by Oset coefficients that depend on density and πN kinematics.

7.4 Cross sections

The total, quasi-elastic, charge-exchange, and absorption cross sections are decomposed into s-wave and p-wave contributions, with isospin-dependent coefficients for each pion charge state.

The fate fractions are:

$$f_{\mathrm{abs}} = \sigma_{\mathrm{abs}} / \sigma_{\mathrm{tot}}, \tag{29}$$

$$f_{\mathrm{CEx}} = \sigma_{\mathrm{CEx}} / \sigma_{\mathrm{tot}}, \tag{30}$$

$$f_{\mathrm{elas}} = 1 - f_{\mathrm{abs}} - f_{\mathrm{CEx}}. \tag{31}$$

8 Post-FSI Processing

8.1 Nucleon selection

After GENIE's `ProcessEventRecord()` returns, the stable descendants of the transported nucleon (slot 3) are examined. The **highest-energy nucleon** among them is selected as the outgoing lead (or recoil) nucleon. Its PDG code may have changed (charge exchange), and its momentum is modified by the cascade.

8.2 Secondary collection

All other stable descendants are stored as `FSISecondary` objects, tagged with:

- `parentRole`: 0 (from lead transport) or 1 (from recoil)
- `pdg`: particle type
- `p4`: 4-momentum
- `rescatterCode`: GENIE's internal fate code (for reweighting)

8.3 Residual nucleus

The residual nucleus used for transport is computed by removing the SRC pair from the full nucleus:

$$Z_{\rm res} = Z - n_p, \qquad A_{\rm res} = A - 2, \tag{32}$$

where n_p is the number of protons in the pair. For ultra-light residuals ($A_{\rm res} < 3$), the full nucleus is used instead to avoid pathological density profiles.

8.4 Pauli blocking (post-FSI)

After both nucleons have been transported, a final Pauli-blocking check is applied:

$$\text{if} \quad |\boldsymbol{p}'_{\rm lead}| < k_F \quad \text{or} \quad |\boldsymbol{p}'_{\rm rec}| < k_F, \quad \Longrightarrow \quad w = 0. \tag{33}$$

Default Fermi momentum: $k_F = 0.220\,\text{GeV}/c$ (tunable via `SetFSITuning`).

8.5 Event classification

Each event is classified by:

- **Charge exchange**: lead or recoil changed PDG ($p \leftrightarrow n$)
- **Elastic-like**: momentum changed, no charge exchange, no non-nucleon secondaries produced
- **Pion counts**: n_{π^+}, n_{π^-}, n_{π^0} tallied separately for lead and recoil transports

9 Full Pipeline Summary

For each Monte Carlo trial:

1. **GCF vertex**: Select pair type, sample CM and relative momenta, compute electron vertex, evaluate σ_{eN} (identical to `genQE`).
2. **Position sampling**: Draw $\boldsymbol{x}_{\rm SRC}$ from $P(r) \propto \rho^2(r)\, r^2$ via rejection sampling.
3. **Compute residual nucleus**: $A_{\rm res} = A - 2$, $Z_{\rm res} = Z - n_p$.
4. **Transport lead nucleon**: Build GENIE event record with nucleon at $\boldsymbol{x}_{\rm SRC}$; run `ProcessEventRecord()` extract highest-energy nucleon and secondaries.

5. **Transport recoil nucleon**: Same procedure, same starting position, same residual nucleus.

6. **Classify**: Detect charge exchange, elastic-like events, count pions.

7. **Pauli blocking**: Reject event if either final nucleon momentum $< k_F$.

8. **Store**: Write event to ROOT tree if $w > 0$.

10 Configuration and Tuning Parameters

Parameter	Default	Setter	Description
FSI model	hN	`SetFSIModel(kHN2018)`	hN or hA cascade
FSI on/off	on	`EnableFSI(true)`	Bypass cascade
k_F	220 MeV/c	`SetFSITuning(220)`	Pauli blocking threshold
δr	0.05 fm	(GENIE internal)	Cascade step size
R_0	1.4 fm	(GENIE internal)	Nuclear radius coefficient
f_{NR}	3.0	(GENIE internal)	Tracking radius multiplier
δR_π	$\sim$1.5	(GENIE config)	Pion de Broglie ring
δR_N	$\sim$1.5	(GENIE config)	Nucleon de Broglie ring
NN medium corr.	on	`INUKE-XsecNNCorr`	Pandharipande–Pieper
Oset model	on	`HNINUKE-UseOset`	Low-energy pion cross sections

GENIE configuration is loaded from XML files in `config/genie_override/` (e.g., `HNIntranuke2018.xml`, `Messenger.xml`).

11 References

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